



Fig. 3: Cell-tower antenna array (Courtesy: antenna-theory.com)

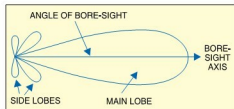


Fig. 4: Typical antenna radiation pattern with side lobes

The gain of an antenna with losses is given by:

$$G = \frac{4\pi A}{\lambda^2}$$

where 'η' is aperture efficiency, 'A' physical aperture area and 'λ' wavelength.

Note that the gain is proportional to the aperture area and inversely proportional to the square of the wavelength. So if the frequency is doubled, or the wavelength halved, the aperture could be decreased four times to maintain the same gain.

## Radiation patterns of various antennas

According to ANITA's website ([www.phys.hawaii.edu/~anita/new/papers/militaryHandbook/freqphas.pdf](http://www.phys.hawaii.edu/~anita/new/papers/militaryHandbook/freqphas.pdf)), radiation patterns of the antennas presented in the previous section are for the most commonly used antenna geometries. The antenna should be viewed as a matching network that takes the power from a transmission line (50-ohm, for example) and matches it to the free space impedance of 377

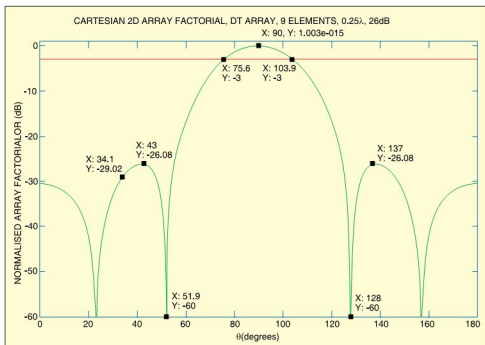


Fig. 5: Cartesian plot

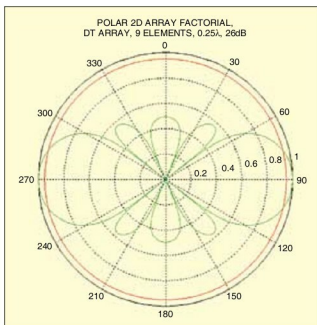


Fig. 6: Polar plot

ohms. The most critical parameter is the change of VSWR with frequency. The pattern usually does not vary much from acceptable to the start of unacceptable VSWRs (>2:1). For a given physical antenna geometric size, the actual radiation pattern varies with frequency.

## Antenna arrays

According to AntennaTheory.com, an antenna array, often called a 'phased array,' is a set of two or more antennas, as shown in Fig. 3. The

signals from the antennas are combined or processed in order to achieve improved performance over a single antenna.

It is generally noticed that the antenna array factors for arrays with uniform weights have unequal sidelobe levels (refer Fig. 4). Often, it is desirable to reduce the highest sidelobes at the expense of increasing number of lower sidelobes. The optimal sidelobe level for a specified beamwidth will occur when the sidelobes are all equal in magnitude.

There is a method for obtaining weights for uniformly spaced linear arrays steered to broadside ( $\theta_0=90$  degrees). It is a popular weighting method because the sidelobe level can be specified, and the minimum possible null-null beamwidth obtained.

Figs 5 and 6 show cartesian and polar plots of the antenna radiation pattern, respectively. Effective selection of  $\lambda$  and the number of elements can reduce the sidelobe level of antenna.